

Linear Regression with One Regressor: Hypothesis Tests and Confidence Intervals

SW Chapter 5

ECON3500: Econometrics and Applications

Spring 2026

Learning Objectives

By the end of this chapter, you will be able to:

- Create hypotheses about slope coefficients and test them using $\hat{\beta}_1$ and its standard error
- Correctly interpret the results of hypothesis tests
- Calculate confidence intervals for β_1
- Take **binary regressors** in stride (and interpret them correctly)
- Understand the implications of **heteroskedasticity** and correct your standard errors
- Know and apply the **Gauss–Markov** theorem to understand the circumstances under which OLS is **BLUE**

Where We Are Going

We want to learn about the slope of the population regression line. We have data from a sample, so there is sampling uncertainty.

- State the population object of interest
- Provide an estimator of this population object
- Derive the sampling distribution of the estimator (large- n normal by the CLT)
- Find the standard error (SE) of the estimator
- Construct t -statistics and confidence intervals

5.1 Testing Hypotheses About One Regression Coefficient

The Sampling Distribution of $\hat{\beta}_1$

Under the Least Squares Assumptions, for n large, $\hat{\beta}_1$ is approximately distributed:

i Note

$$\hat{\beta}_1 \sim N \left(\beta_1, \frac{\sigma_v^2}{n(\sigma_X^2)^2} \right), \quad \text{where } v_i = (X_i - \mu_X)u_i$$

Note: We won't compute variances by hand, but the intuition is useful.

Hypothesis Testing: General Setup

- Null hypothesis and **two-sided** alternative:

i Note

$$H_0 : \beta_1 = \beta_{1,0} \quad \text{vs} \quad H_1 : \beta_1 \neq \beta_{1,0}$$

- Null hypothesis and **one-sided** alternative:

i Note

$$H_0 : \beta_1 = \beta_{1,0} \quad \text{vs} \quad H_1 : \beta_1 < \beta_{1,0}$$

General Approach

- In general:

Note

$$t = \frac{\text{estimator} - \text{hypothesized value}}{SE(\text{estimator})}$$

- For testing $\bar{Y}$: $t = \frac{\bar{Y} - \mu_{Y,0}}{s_y / \sqrt{n}}$
- For testing β_1 :

Note

Testing $H_0 : \beta_{1,0} = 0$

- Construct your t -statistic:

 **Note**

$$t = \frac{\hat{\beta}_1 - \beta_{1,0}}{SE(\hat{\beta}_1)}$$

- Reject at level α if $|t| > c_{\alpha/2}$
- In practice, almost always two-tailed tests
- Relies on large- n normal approximation (rule of thumb: $n > 30$)

Common critical values:

Choosing a Significance Level

- We fix α as the **significance level** (Type I error)
- α is the probability of falsely rejecting the null
- Typical choice: $\alpha = 0.05$ (5% false positives)
- Is that too high? Why not make α super small?

Choosing a Significance Level (Tradeoff)

- Smaller α makes it harder to reject H_0
- Fewer false positives, but more false negatives
- **Power** = $1 - \beta$ (probability of rejecting a false null)
- Tradeoff between significance (α) and power (β)

Conducting a Hypothesis Test

Is mother's education associated with birthweight?

```
{{< include stata/inference.log >}}
```

Testing $H_0 : \beta_1 = a$

- More general hypotheses:
 - $H_0 : \beta_1 = a$
 - Adjust the t -statistic

 **Note**

$$t = \frac{\hat{\beta}_1 - a}{SE(\hat{\beta}_1)}$$

Economic vs. Statistical Significance

- If statistically significant, examine magnitude
- Statistical significance $\neq$ economic importance
- Wrong sign? Model may be misspecified
- If insignificant, you *might* exclude—but be cautious with small samples

5.2 Confidence Intervals for β_1

Confidence Intervals for β_1

Recall, a 95% confidence interval is:

- The set of values that cannot be rejected at the 5% significance level
- A set-valued function of the data that contains the true parameter 95% of the time

Because the t -statistic is approximately $N(0, 1)$ in large samples:

 **Note**

$$95\% \text{ CI for } \beta_1 : \quad \hat{\beta}_1 \pm 1.96 SE(\hat{\beta}_1)$$

5.3 Regression When x Is Binary

Regression When X Is Binary

Sometimes a regressor is **binary**:

- $X = 1$ if small class size, $X = 0$ otherwise
- $X = 1$ if female, $X = 0$ if male
- $X = 1$ if treated, $X = 0$ otherwise

Binary regressors are called **dummy variables**.

Gender is not binary, but it is recorded as binary in many datasets—data availability shapes our understanding of the world.

Interpreting a Binary X (1)

Population model: $Y_i = \beta_0 + \beta_1 X_i + u_i$

When $X_i = 0$:

- $Y_i = \beta_0 + u_i$
- $E[Y_i \mid X_i = 0] = \beta_0$

When $X_i = 1$:

- $Y_i = \beta_0 + \beta_1 + u_i$
- $E[Y_i \mid X_i = 1] = \beta_0 + \beta_1$

Interpreting a Binary X (2)

Therefore:

 **Note**

$$\beta_1 = E(Y_i \mid X_i = 1) - E(Y_i \mid X_i = 0)$$

So β_1 is the population difference in group means.

Interpreting a Binary X : Stata Output

Is sex associated with birthweight?

```
{{< include stata/binary.log >}}
```

Interpreting a Binary X (Example)

Is sex associated with birthweight?

$$\hat{bwght} = 117.17 + 2.94 \text{ male}$$

- Average birthweight of female babies: $E[bwght \mid \text{male} = 0] = 117.17$ ounces
- Average birthweight of male babies:
 $E[bwght \mid \text{male} = 1] = 117.17 + 2.94 = 120.11$ ounces

5.4 Heteroskedasticity and Homoskedasticity

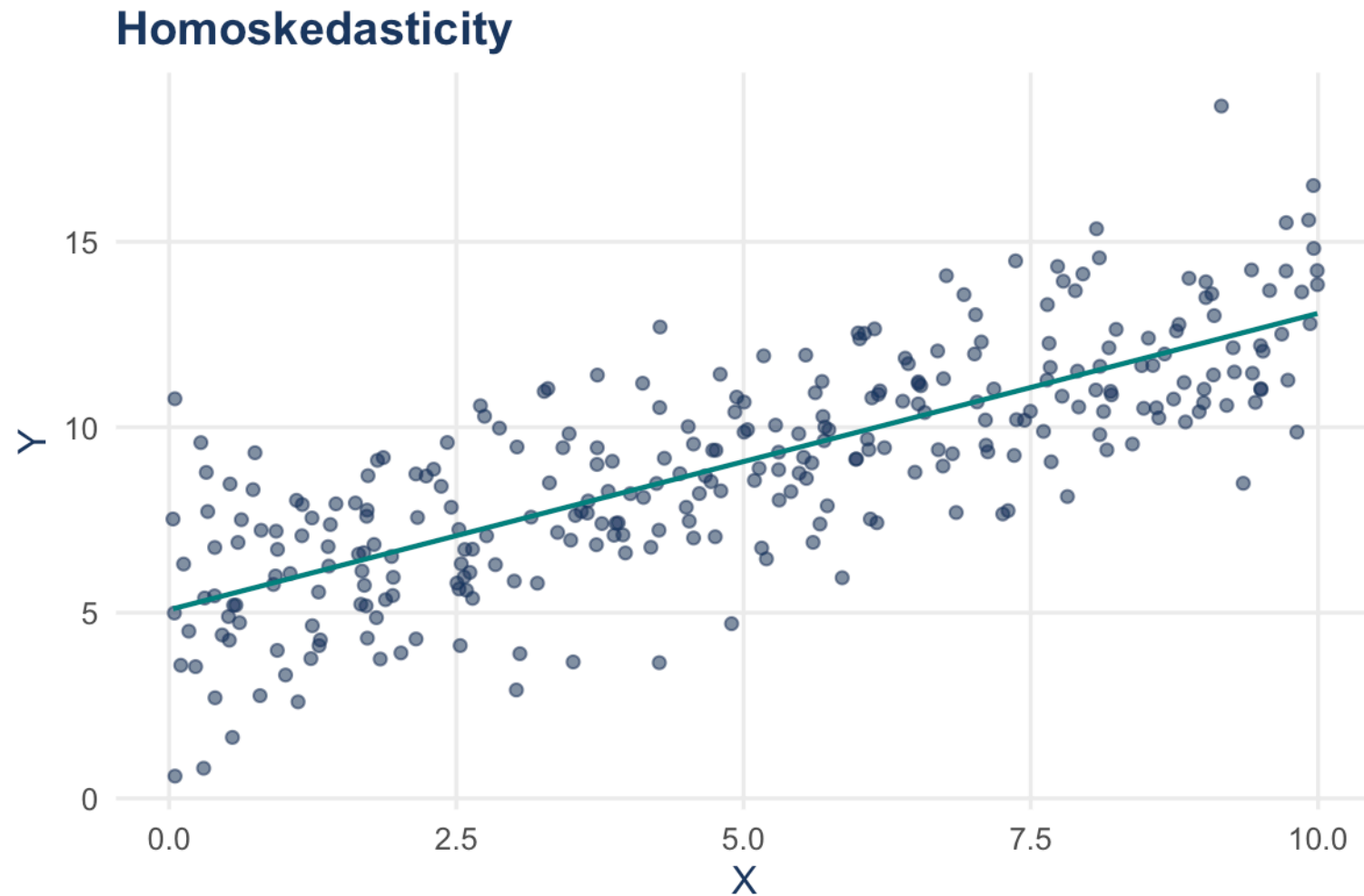
Heteroskedasticity and Homoskedasticity

1. What do the terms mean?
2. Consequences of heteroskedasticity
3. Implications for computing standard errors

If $\text{var}(u \mid X = x)$ is constant, u is **homoskedastic**.

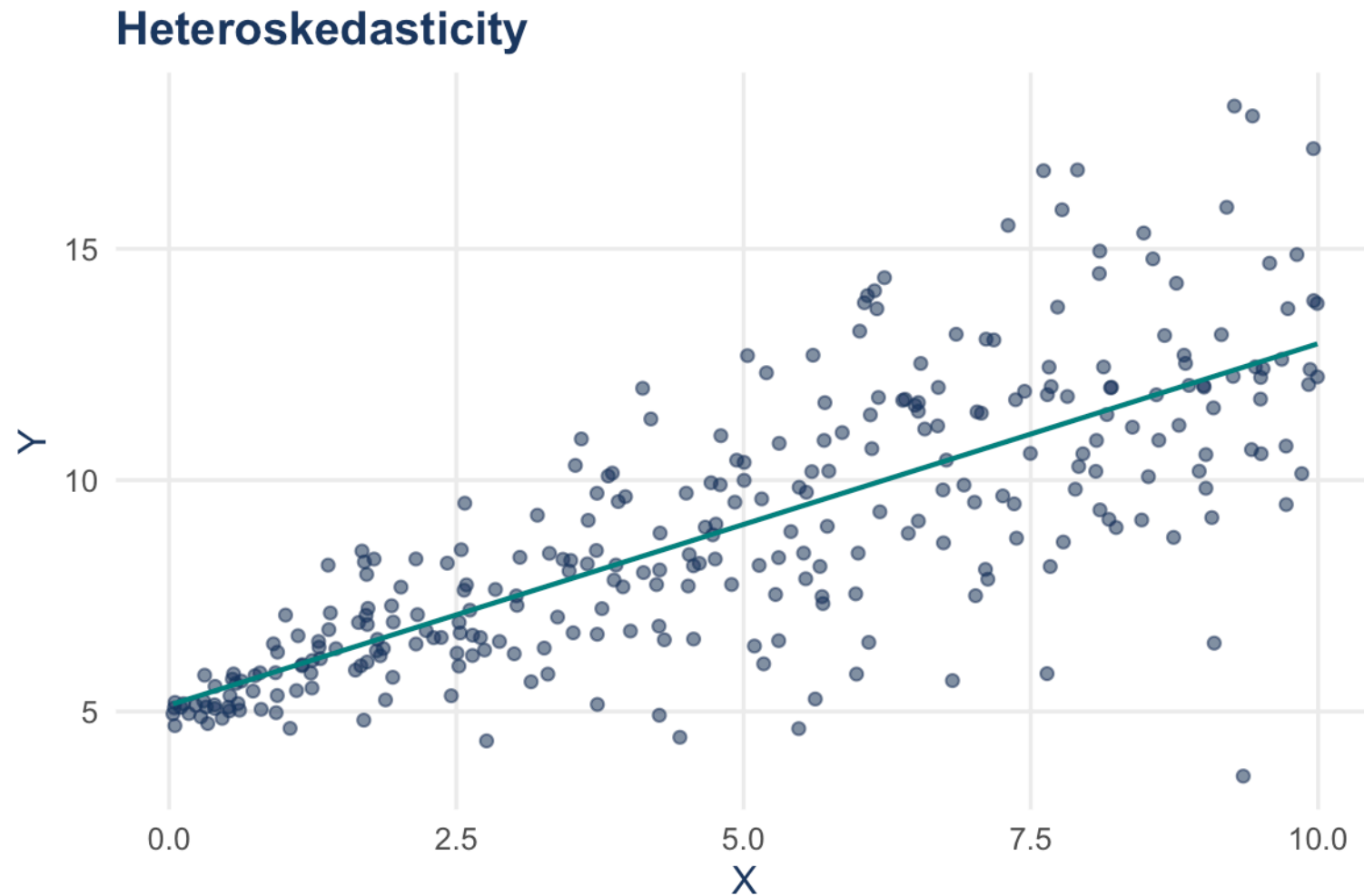
Otherwise, u is **heteroskedastic**.

Homoskedasticity in a Picture



- The variance of u is constant
- The variance of u does not depend on X

Heteroskedasticity in a Picture



- The variance of u is not constant

- The variance of u **does** depend on X

Does Heteroskedasticity Affect $\hat{\beta}_1$?

Recall the least squares assumptions:

1. $E(u \mid X = x) = 0$
2. (X_i, Y_i) are i.i.d.
3. Large outliers are rare

Heteroskedasticity concerns $\text{var}(u \mid X = x)$.

Because we did **not** assume homoskedasticity, we have implicitly allowed heteroskedasticity.

So Who Cares?

Heteroskedasticity does **not** affect point estimates of β_1 .

But it **does** affect your standard errors.

Homoskedastic-only standard errors are unbiased only under homoskedasticity. We adjust using **heteroskedasticity-robust** standard errors.

Heteroskedasticity-Robust Standard Errors (Stata)

```
{{< include stata/homo1.log >}}
```

Heteroskedasticity-Robust Standard Errors (Stata)

```
{{< include stata/homo2.log >}}
```

Heteroskedasticity: The Bottom Line

- If errors are homoskedastic or heteroskedastic **and** you use robust SEs, you're OK
- If errors are heteroskedastic and you use homoskedastic-only SEs, your SEs are wrong
 - Could be too big or too small
- The two formulas coincide (large n) only under homoskedasticity
- So: **always use heteroskedasticity-robust SEs**

5.5 Gauss–Markov Theorem

The Extended Least Squares Assumptions

Consider our three LS assumptions (needed for unbiasedness):

1. $E(u \mid X = x) = 0$
2. (X_i, Y_i) are i.i.d.
3. Large outliers are rare

Plus one more:

4. u is **homoskedastic**

Gauss–Markov Theorem

Under these four extended LS assumptions, $\hat{\beta}_1$ has the smallest variance among **all linear estimators**.

This is the **Gauss–Markov Theorem**.

Under GM, OLS estimators are **BLUE**:

- Best
- Linear
- Unbiased
- Estimators

OLS Limitations

- Homoskedasticity often doesn't hold
- GM is only for **linear** estimators (a small subset)
- If we know the form of heteroskedasticity, we can use **weighted least squares**
- With many outliers, **least absolute deviations (LAD)** can be more efficient

In most applied regression analysis, we use OLS—so that is what we will do, too.

Conclusion

Key ideas to remember:

- Hypothesis tests and confidence intervals for β_1
- Interpreting binary regressors
- Why heteroskedasticity matters for standard errors
- The Gauss–Markov conditions and what BLUE means

Practice Problems

Practice Problem (S&W-style): Tests and CI

An OLS regression reports $\hat{\beta}_1 = 1.5$ with $SE(\hat{\beta}_1) = 0.4$. The sample size is $n = 60$.

- Test $H_0 : \beta_1 = 1$ at the 5% level (two-sided)
- Compute the 95% CI for β_1

Practice Problem: Walkthrough

- $t = (1.5 - 1)/0.4 = 1.25$ with $df = 58$
- Two-sided p -value $= 2 P(T_{58} > |1.25|)$
- 95% CI: $1.5 \pm t_{0.975,58} \times 0.4$

Stata:

```
display (1.5-1)/0.4  
display 2*ttail(58,abs((1.5-1)/0.4))  
display 1.5 - invttail(58,0.025)*0.4  
display 1.5 + invttail(58,0.025)*0.4
```